Literature status: the superreflexive non-summing lineability problem

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22 July 2026

Status: literature already answered. Kitson–Timoney (2011), Theorem 4.1, explicitly answers the problem and proves a stronger simultaneous spaceability theorem.

Source question

Botelho, Diniz, and Pellegrino quote as Problem 1.1 the following question of Puglisi and Seoane-Sepúlveda [1]: if E is an infinite-dimensional superreflexive Banach space and $p \geq 1$, is

$$\mathcal{L}(E;F) \setminus \Pi_p(E;F)$$

lineable for every infinite-dimensional Banach space F? Their Theorem 2.1 answers this when either E contains a complemented infinite-dimensional subspace with an unconditional basis or F contains an unconditional basic sequence. Their Remark 2.2 leaves precisely the case where both structural conditions fail.

The source location is arXiv:0811.0092v5, Introduction, Problem 1.1 and Section 2, Theorem 2.1 and the following remark (PDF pages 1–4).

Decisive later theorem

Kitson and Timoney explicitly write that their Section 4 improves the earlier results and answers the original problem [2]. Their Theorem 4.1 states: if X and Y are infinite-dimensional Banach spaces and X is superreflexive, then

$$\mathcal{K}(X,Y) \setminus \bigcup_{1 \leq p < \infty} \Pi_p(X,Y)$$

is spaceable as a subset of $\mathcal{K}(X,Y)$.

This is Theorem 4.1 on accepted-manuscript pages 7–8 (published article pages 680–686, DOI below).

Why this settles the source question

Spaceability is stronger than lineability. Theorem 4.1 supplies a closed infinite-dimensional subspace $W \subset \mathcal{K}(E,F)$ such that every nonzero $T \in W$ fails to be absolutely p-summing for *every* finite p. For each fixed $p \geq 1$,

$$W \setminus \{0\} \subseteq \mathcal{K}(E,F) \setminus \Pi_p(E,F) \subseteq \mathcal{L}(E,F) \setminus \Pi_p(E,F).$$

Thus the answer is affirmative for all pairs (E, F) in the source problem, including its simultaneous pathological case. No part of that problem remains open. The supporting authors knew they were answering it; this is not merely an agent-identified implication.

Evidence and local artifacts

The original paper is stored locally as `source_paper.pdf`. The accepted supporting manuscript was inspected through Trinity College Dublin's repository. Direct automated downloads returned an anti-bot HTML challenge rather than a PDF, so no false `.pdf` artifact is retained; the stable DOI, repository link, and exact theorem location are recorded in `supporting_paper_access.md`.

References

[1] G. Botelho, D. Diniz, and D. Pellegrino, *Lineability of the set of bounded linear non-absolutely summing operators*, J. Math. Anal. Appl. 357 (2009), 171–175; arXiv:0811.0092v5. `https://arxiv.org/abs/0811.0092`

[2] D. Kitson and R. M. Timoney, *Operator ranges and spaceability*, J. Math. Anal. Appl. 378 (2011), 680–686. `https://doi.org/10.1016/j.jmaa.2010.12.061`